Congestion control

1. What is congestion control? Discuss explicit & implicit congestion control and their difference. Discuss the technique of TCP congestion control.

Congestion is a state occurring in network layer when the message traffic is so heavy that it slows down the network response time. The technique to control this phenomenon is called congestion control.

In implicit signaling, there is no communication between the congested nodes and the source. The source guesses that there is a congestion somewhere in the network.

In explicit signaling, the node that experiences congestion can explicitly send a signal to the source or destination.

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Congestion Policy in TCP :

1. Slow start phase : starts slowly increment is exponential to threshold.

2. Congestion Avoidance Phase : After reaching the threshold ~~in~~ increment is by 1.

3. Congestion detection Phase : Sender goes back to slow start Phase or Congestion avoidance phase.

Slow start phase : exponential increment — In this phase after every RTT the congestion window size increments exponentially.

Congestion avoidance Phase : additive increment — this phase starts after the threshold value.

Congestion detection Phase : multiplicative decrement — If congestion occurs, the congestion window size decreases. the only way a sender can guess that congestion has occured is the need to retransmit a segment. Retransmission is needed to recover a missing Packet which is assumed to have been dropped by a router due to congestion. Retransmission can occur in one of two cases : when the RTO timer times out or when three duplicate ACKs are recieved.

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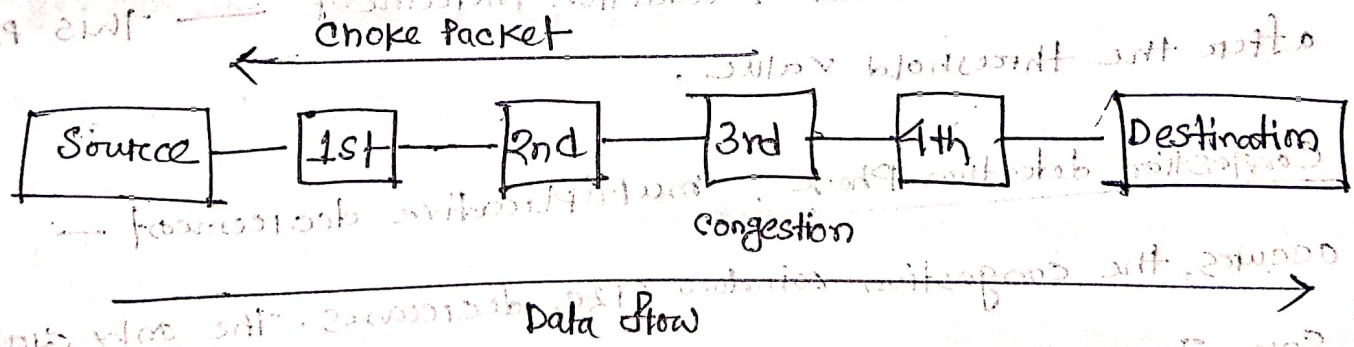
2. Discuss hop-by-hop choke Packet technique.

It is applicable to both virtual networks as well as datagram Subnets. A choke Packet is a Packet Sent by a node to the Source to inform it of congestion. Each router monitors its resources and the utilization at each of its output lines. Whenever the resource utilization exceeds the threshold value which is set by the administrator, the router directly sends a choke Packet to the source giving it a feedback to reduce the traffic. The intermediate nodes through which the Packets has traveled are not warned about congestion.



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3. **Leaky Bucket :** The leaky bucket algorithm is a method of temporarily storing a variable number of request and organizing them into a set-rate output of packets in an asynchronous transfer mode (ATM).

Steps :

1. When host wants to send packet, packet is thrown into the bucket.
2. The bucket leaks at a constant rate, meaning the network interface transmits packets at a constant rate.
3. Bursty traffic is converted to a uniform traffic by the leaky bucket.
4. In practice the bucket is a finite queue that outputs at a finite rate.

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4. **Token Bucket:** The token bucket is an algorithm used in packet switched computer networks and telecommunications networks.

Steps :

1. In regular intervals tokens are thrown into the bucket.
2. The bucket has a maximum capacity.
3. If there is a ready packet, a token is removed from the bucket and the packet is sent.
4. If there is no token in the bucket, the packet can't be sent.

5. Short note on load shedding.

It is a technique used in information systems, especially web services, to avoid overloading the system and making it unavailable for all users. The idea is to ignore some request rather than crashing a system and making it fail to serve any request.

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